Phase 1: Assisted Development

Module 1: Foundations of AI Collaboration

Building the mindset and skills to effectively collaborate with AI assistants

👋 Agenda

- Introductions
- On the topic of AI
- Learning objectives
- How LLMs work (brief)
- The Relationship with your assistant
- Managing the human dynamic
- Model capabilities & limitations
- Balancing control & delegation
- Demonstration & our Phase 1 project

👋 Introductions

- Name
- Experience with AI coding tools
- What you're most curious about

🤖 On the Topic of AI

- In general
- In software development
- So then what?

💡 Our Perspective on AI Technology

Realistic:

- Excited about the benefits and precision potential
- Realistic about energy costs and computing requirements
- It's a powerful, fun, and technically rich set of innovations
- Change, especially when it comes fast, can be destablising for people

Our Interest:

- Model selection over defaulting to the most capable/expensive
- Practical domain-specific applications over AGI/ASI speculation
- Interest in local and specialized models

💡 Our Perspective on AI in Software Development

What motivates our belief in this foundational shift:

- The output from AI assistants is **too good to dismiss**
- The sense of productivity gains during implementation is **profound**
- We know we can **increase precision** by applying higher quality context
- The quality of the models is getting **noticeably better, very quickly**

🤔 What to Focus On

Flow

- When you get this, everything else falls in place
- Identify friction and mitigate it
- Flow == intent $\propto$ result

Learning

- Prioritise your learning journey
- Either continuous improvement or increased understanding

Quality

- What you do with the time & space your AI assistant gives you back

📚 Learning Objectives

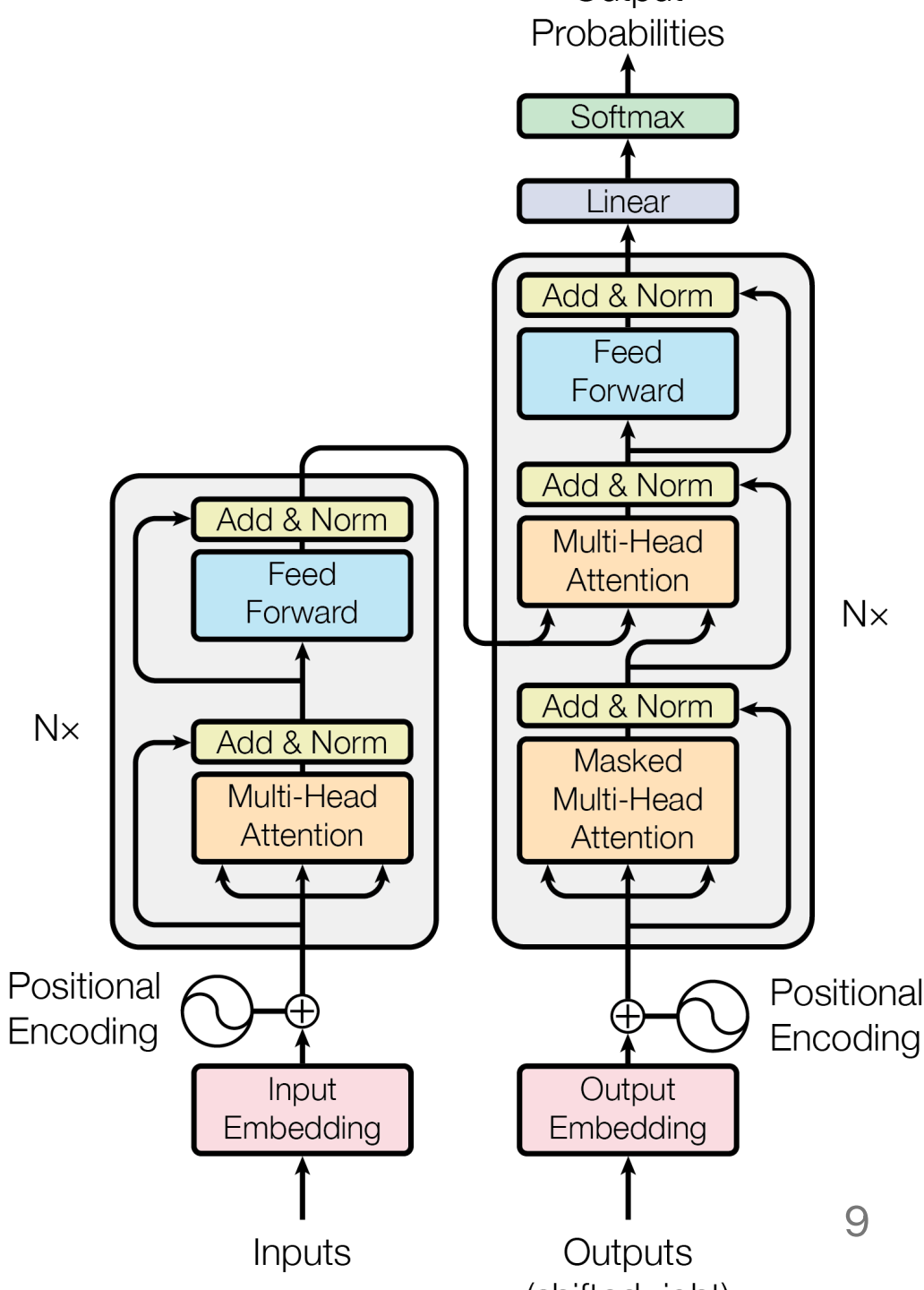
- **Goal**: How to collaborate with an AI assistant to build working code and docs
- **Takeaway**: Chosen tool, model, and prompt and process to create a service/client combo with expected results
- **Hope**: You will have lots of feels 🥺 while remaining curious

Along the way you will:

1. **Apply different personas** while working with your assistant
2. **Experiment** with various coding tools and models
3. **Craft effective prompts** for planning, code generation, exploration, problem-solving, and learning
4. Recognize AI's **capabilities** and **limitations** with various tasks
5. Balance **control and delegation** when working with AI

How LLMs Work (a discussion)

- Tokens
- Training
- Parameters
- Inference
- Attention
- Context window
- Coding tools



👤 AI Roles & Personas

AI assistants act differently depending on the **role** you give them

Role	Purpose
Coder	Writes code from context
Planner	Structures tasks
Architect	Designs systems
UI Designer	Handles layout/UX
Skeptic	Finds flaws and holes
Mentor	Provides explanations

🧠 The *Relationship* with your Assistant

- It *feels* like you're working with a human
  - Familiar, easeful, flexible, using *our* language
- You're just filtering for a *close enough* sequence of tokens
  - Who are you getting upset with?
- Prioritise **yourself** in these interactions
  - Remain kind and curious

🥺 Managing the Human Dynamic

Emotional Challenges

- Frustration with repeating imprecise results
- Overwhelm from too many possibilities
- Loss of control and agency
- Uncertainty about quality

Strategies

- Plan first, start small, and build your capacity
- Maintain clear boundaries (scope, direction, priority)
- Practice "non-attachment" to AI output (Git is your friend)
- Stay curious, experiment, and move towards flow (mitigating friction)

Capabilities & Limitations

> Really, the point of modules 1-3 is to help you build this intuition

But here are a few things we've noticed

🏆 Current AI is good at

- Code generation and boilerplate (esp when given an example)
- Refactoring, pattern matching, consistency checks, and translation
- Documentation, explanation, and dictation

💩 Common failure modes

- Context window limitations
- Hallucinations (e.g. non-existent APIs)
- Limited business domain understanding

🕹️ Balancing Control & Delegation

Maintain Control Over

- Architectural decisions
- Business logic validation
- Quality standards
- Final implementation choices

Delegate to AI

- Code scaffolding and boilerplate
- Documentation generation
- Test case suggestions

Key: You remain the decision-maker and are ultimately responsible

A Demonstration

- Model API Keys
- OpenRouter setup
- Tool and model choice
- See how context matters

🪜 Project Progression: Module 1

Our Initial Config Service

- Collaborating on the initial plan
- Basic CRUD API for configuration values
- Simple data persistence and validation
- API documentation and testing fundamentals
- Web UI administration interface for managing configurations
- Initial experiences of flow friction, learning capture, and quality adherence
- Satiate our appetite for code gen (even though that isn't *really* the priority)
- Begin to establish a consistency-inducing workflow

🚀 Let's Go!

Remember:

- **You are *still* ultimately responsible/accountable**
- **Enage the many personas of your assistant**
- **Focus on learning the collaboration patterns**
- **Non-attachment to initially generated output**
- **Be aware of *lazy* tendencies**

Answer: How can my assistant help me to optimise **flow, learning, and quality**?

First step: Tool and model selection, and project setup